Agroforestry SmartAgroforst: Poplar biomass modelling

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Abstract

Short-rotation agroforestry systems are increasingly promoted as a low-carbon feedstock option for bio-based value chains, but their economic viability depends on coordinated decisions on stand establishment, multi-cycle harvesting, and product cascading in spatially distributed supply chains. This paper develops a mixed-integer linear programming (MILP) model for the integrated design and operation of an agroforestry biomass supply chain with explicit age-lag constraints that link consecutive harvests at each site. The formulation couples allometric yield functions for different biomass types with a multi-product, multi-period network design model including storage, processing, and product quality cascades. The model is applied to a stylised case study of Central European poplar-based alley-cropping systems to explore trade-offs between harvest rotation length, product portfolios, and infrastructure configuration. We outline a computational experimentation plan to test the impact of allometric parameter uncertainty, price scenarios, and policy incentives on the optimal deployment of agroforestry systems in cleaner bioeconomy pathways.

Keywords: Agroforestry, Short-rotation coppice, Biomass supply chain, Product cascading

1. Introduction

Agroforestry systems that integrate trees with agricultural crops (silvoarable systems) are increasingly recognised as a promising land-use option to reconcile climate mitigation, biodiversity conservation, and rural income generation on the same hectare of land [14].

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Temperate agroforestry systems (AFS) based on fast-growing species such as poplar, willow, alder, and black locust can deliver substantial biomass yields while improving soil quality and providing multiple ecosystem services compared to conventional monocultures [7]. Biomasses produced in AFS is considered as a sustainable feedstock and can be used either energetically, chemically or materially (e.g. in the building sector). The latter two options are generally preferred over the energetic usage from a cleaner production perspective as biogenic carbon is stored in products during their life span which contributes to climate change mitigation.

Despite substantial research on ecological advantages and pioneering initiatives to establish AFS in practice, the share of AFS in agricultural production systems in still small. E.g. in Germany about 1200 hectars of silvoarable AFS are established in 2025, which accounts for about 0.02% of total agricultural land use [DEFAF]. One of the core challenges in establishing AFS systems in large scale in practice, is a lack of opportunities to generate substantial revenues from woody biomasses of AFS. On the other side, the rise of bioeconomy creating substantial demands of woody biomass requires a reliable and cost-efficient feedstock supply potentials. As woody biomasses are bulky feedstocks with low energy density, the design of cost-efficient supply chains is crucial to ensure the economic viability of AFS-based value chains. This requires coordinated decisions on stand establishment, multi-cycle harvesting, and product cascading in spatially distributed supply chains, which can be supported by integrated optimisation models that link stand-level growth dynamics to regional supply chain design.

In the last two decades, a large body of research has analysed biomass supply chains for bioenergy and biorefineries using mathematical programming models, often in the form of mixed-integer linear programming (MILP) [16, 12]. These models typically optimise network configuration, feedstock sourcing, and logistics under cost or profit objectives. They often treat biomass availability as an exogenous input or rely on simplified agricultural production decisions. Moreover, downstream processing of biomasses is concentrated on one particular value chain e.g. bioenergy plants consuming all produced biomasses. Recent work has started to integrate operational details such as various biomass types, biomass growth models and regeneration processes in multi-period supply chain models.

The present paper contributes to this literature in four ways. First, we propose a novel MILP formulation for AFS-based biomass supply chain design that explicitly links establishment decisions and consecutive harvest decisions, ensuring that harvesting decisions at each site are consistent with minimum and maximum rotation ages. Second, we embed allometric yield functions for different biomass types generating age-specific yields for different woody biomass types (like stems, branches, or leaves and twigs). Third, these different woody biomass types can be processed in specific downstream industries, like chemical or paper industry as well as energy sector. The model aims to provide strategic decision support for establishing regiona .AFS-based bioeconomy ecosystems connecting multiple industries and product markets.

The remainder of the paper is structured as follows. Section @ref(sec:litreview) reviews the relevant literature on agroforestry biomass systems, allometric biomass modelling, and biomass supply chain optimisation. Section @ref(sec:allometrics) introduces the allometric models adopted to represent biomass growth of different tree species and biomass types. Section @ref(sec:milp) summarises the MILP formulation for the agroforestry supply chain with age-lag constraints. Section @ref(sec:experiments) outlines the planned computational experiments for the case study and discusses expected insights for cleaner production strategies.

2. Literature review

Agroforestry biomass systems, tree and stand biomass models, and biomass supply chain optimisation have all been studied extensively, but they are rarely combined in an integrated modelling framework that links age-dependent stand dynamics to regional product-cascading value chains [16, 12, 5]. This section reviews (i) agroforestry and short-rotation systems with emphasis on yields and economics, (ii) allometric and age-dependent biomass yield models, and (iii) biomass supply chain optimisation and cascading use of biomass, and concludes by identifying the modelling gap addressed in this paper.

2.1. Biomass supply chain optimisation

Biomass supply chain design and planning have been widely studied using MILP models at strategic, tactical, and operational levels Zahraee et al. [16]; Sharma et al.

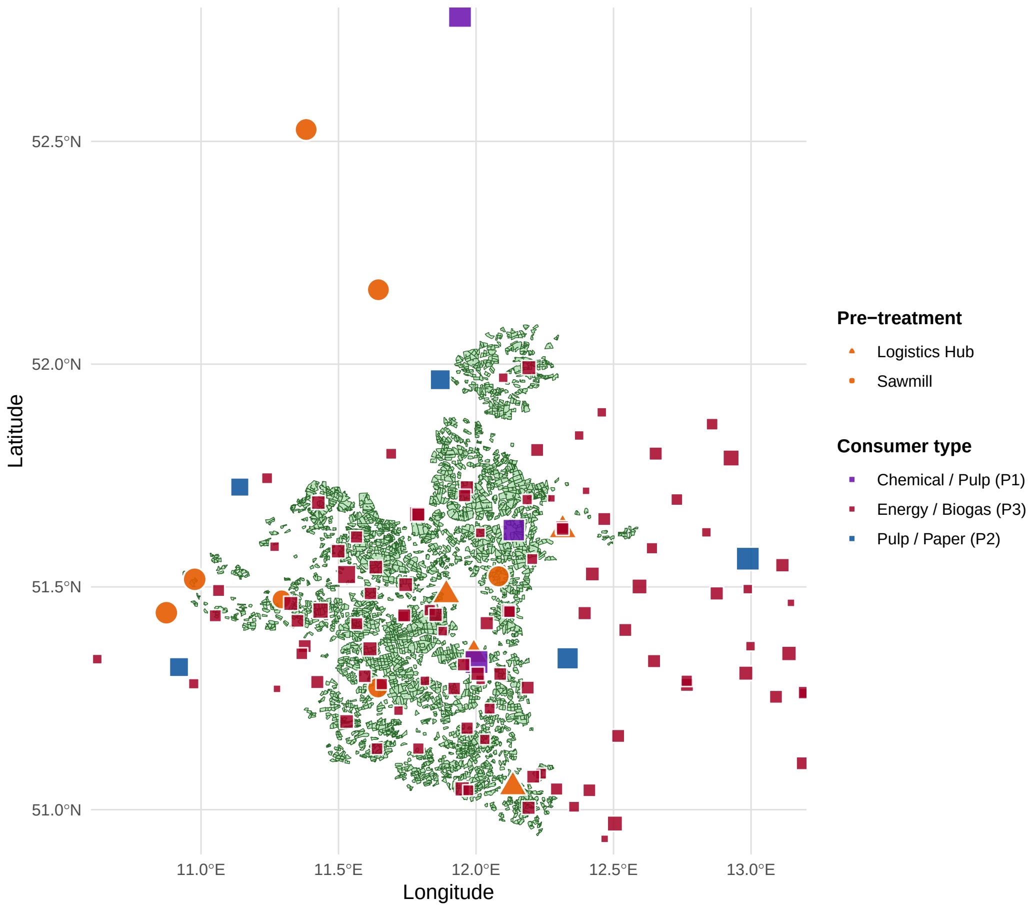

Figure 1: AFS supply chain network in Saxony-Anhalt. Green polygons: potential KUP sites. Orange circles: pre-treatment facilities (sawmills and logistics hubs, radius proportional to processing capacity). Coloured squares: consumer sites (violet: chemical/pulp, blue: pulp/paper, red: energy/biogas).

[12]; Espinoza et al. [5]. Comprehensive reviews classify models according to decision scope (location, capacity, routing, inventory), time horizon, uncertainty treatment, and sustainability metrics, and show a strong emphasis on bioenergy and biorefinery systems Lamers et al. [8]; Espinoza et al. [5]. Typical models optimise the number, size, and locations of processing facilities, biomass sourcing patterns, multimodal transport flows, and inventory policies under cost- or profit-oriented objectives, sometimes complemented by greenhouse gas emissions or other environmental indicators Eksioglu et al. [4]; Arabi et al. [1].

A key methodological feature of many biomass supply chain models is the integration of long-term location and capacity decisions with multi-period logistics decisions such as harvesting, transport, storage, and processing Eksioglu et al. [4]; Zahraee et al. [16]. For example, mixed-integer formulations have been used to simultaneously determine biorefinery locations, their capacities, biomass collection and storage sites, and time-dependent flows of biomass and biofuel, explicitly accounting for seasonality, deterioration, and inventory trade-offs Eksioglu et al. [4]. These models demonstrate that a significant share of biofuel production costs is attributable to logistics, and that integrated optimisation of network design and operations can substantially reduce costs compared to decoupled decision making Eksioglu et al. [4]; Arabi et al. [1].

However, in virtually all of these models, biomass availability over time is treated as an exogenous input, either as annual supply potentials per region or as simple time profiles based on harvest windows and assumed yields Zahraee et al. [16]; De Meyer et al. [3]. Stand-level growth dynamics, age-class structure, and rotation decisions are not represented explicitly; instead, land-use or yield scenarios enter as fixed parameters De Meyer et al. [3]; Lamers et al. [8]. As a result, these models cannot explore trade-offs between rotation length, stand management, and supply chain configuration in a unified optimisation framework, especially not for agroforestry systems with repeated coppice cycles and strong interactions between tree growth and harvesting decisions.

2.2. Agroforestry and short-rotation systems

Agroforestry deliberately integrates woody perennials with crops and/or livestock on the same management unit, with the aim of enhancing productivity, profitability, and environmental performance Nair [10]; Toensmeier [14]. In temperate regions, short-rotation agroforestry systems (SRAFS) often take the form of alley-cropping, where double or multiple tree rows alternate with crop strips, enabling simultaneous production of agricultural and woody biomass Huber et al. [7]. Empirical trials in Southern Germany and Central Europe show that SRAFS with poplar, willow, black alder, and black locust can achieve mean annual increments (MAI) of roughly 7–10 t per ha and year in aboveground woody biomass during the first rotation, with pronounced species-specific differences in growth trajectories, stand structure, and mortality (see Huber et al. [7]; Trnka et al. [15]).

For short-rotation coppice (SRC) and very short-rotation coppice (vSRC) poplar plantations in Italy and Central Europe, economic analyses indicate that rotations of about 5–7 years at moderate planting densities can be competitive with arable crops, but profitability is highly sensitive to yield assumptions, establishment costs, and logistics [13]. This sensitivity implies to couple stand-level growth and models with supply chain optimisation models, rather than treating biomass availability as an exogenous, time-invariant parameter [16]. Beyond yields and economics, agroforestry and SRC systems have been widely analysed for carbon sequestration, nutrient cycling, and biodiversity, reinforcing their role as candidate feedstock systems for regional bioeconomy strategies that emphasise material-before-energy use of biomass (see e.g. Toensmeier [14]).

Thus, a fundamental cornerpiece of AFS modeling is the representation of biomass growth dynamics, which can be achieved through allometric models that link tree biometric variables to biomass, and through age-dependent stand biomass functions that capture growth trajectories over rotation cycles. These models provide the necessary link between stand structure and biomass availability, which can then be integrated into supply chain optimisation models to explore trade-offs between rotation length, product portfolios, and infrastructure configuration in AFS supply chains.

2.3. Tree-level allometric models

Allometric tree biomass models relate easily measured variables, such as stem base diameter (SBD) or diameter at breast height, to aboveground dry biomass using power-law functions of the form $M = b_0\,\mathrm{SBD}^{b_1}$, with species-specific coefficients b_0 and exponents b_1 [6]. These models allow non-destructive estimation of individual-tree and stand biomass from diameter measurements and have been widely used in forestry and agroforestry for both carbon accounting and management planning [6, 14].

For SRAFS in Southern Germany, species-specific allometric functions based on SBD measured 10 cm above ground were developed for black alder, black locust, poplar (Max 3), and willow (Inger) Huber et al. [6]. The reported models share a common exponent $b_1 = 2.603$ across species, while the scale parameter b_0 ranges from about 0.025 to 0.041, reflecting differences in wood density and branching pattern Huber et al. [6]. Applying these functions to all shoots per tree and aggregating over observed tree densities yields annual aboveground leafless dry biomass at plot and stand level, from which current annual increment (CAI) and MAI curves can be derived for each species and management system Huber et al. [7].

These SRAFS studies demonstrate that stand structure, diameter distributions, and mortality strongly affect aggregate yields, with black locust showing high wood density and strong size inequality and willow showing high shoot densities but low wood density and diameter growth Huber et al. [7]. From a modelling perspective, the key point is that tree-level allometrics provide a robust mapping from biometric state variables to biomass, which can be exploited either to build detailed stand simulators or to construct age-class yield tables for optimisation models Huber et al. [6].

2.4. Age-dependent biomass yield functions

Existing allometric models link tree biomass to diameter or height but do not yet provide **stand-level** biomass as an explicit function of stand age. Age-dependent biomass functions are useful for yield projections in short-rotation forestry and agroforestry, where management is organized in discrete rotations (e.g. 5–15 years). [11, 13]

A common approach is to derive age functions from rotation trials that report mean annual increment (MAI) over time and identify the age at which MAI plateaus. For poplar SRWC in northern Poland, Niemczyk et al. (2021) showed that MAI in above-ground dry biomass increases up to about age 10; beyond this, the MAI curve flattens, suggesting a biologically optimal rotation age around 10 years because the slope of MAI is near zero. [11] Reported MAI values for single-stem poplar plantations in a 10-year rotation ranged approximately from 1–15 Mg DM ha^{-1} yr^{-1} depending on cultivar. [11] Similar magnitudes and rotation lengths (about 5–7 years for SRC, up to 10–15 years for

SRC/SRF) are reported for poplar SRC systems in Italy, where whole-rotation averages of roughly 15 Mg DM ha^{-1} yr^{-1} are considered economically viable. [13]

A simple and widely used parametric form for age-dependent stand biomass $B(t)$ is the Chapman–Richards (or Mitscherlich/monomolecular) function:

$$B(t) = B_{\max}\left(1 - e^{-k \cdot t}\right)^m$$

with stand age t in years, asymptotic biomass $B_{\max}$ (Mg DM ha^{-1}), rate parameter k (year^{-1}), and shape parameter m. This formulation captures the rapid juvenile growth, a phase of approximately linear biomass accumulation, and a saturating phase where growth slows. [11] The corresponding current annual increment (CAI) and mean annual increment (MAI) are

$$\mathrm{CAI}(t) = \frac{dB}{dt}, \qquad \mathrm{MAI}(t) = \frac{B(t)}{t}.$$

The age maximizing MAI (optimal rotation under a volume/biomass criterion) can be derived analytically or determined numerically. Fitting such functions to observed age–biomass or age–volume series from rotation trials allows cultivar-specific yield curves that are directly usable in agroforestry simulation models. [11, 13]

For illustration, we parameterize three stylized poplar curves representing low, medium, and high productivity on temperate mineral soils. Parameter values are loosely informed by the ranges reported by Niemczyk et al. (2021) and Testa et al. (2014) and should be recalibrated when site-specific data are available. [11, 13]

- Low productivity: $B_{\max} = 80$ Mg DM ha^{-1}, $k = 0.18$, $m = 1.1$
- Medium productivity: $B_{\max} = 120$ Mg DM ha^{-1}, $k = 0.20$, $m = 1.2$
- High productivity: $B_{\max} = 160$ Mg DM ha^{-1}, $k = 0.22$, $m = 1.3$

These values yield MAI maxima between about 8 and 12 years with peak MAI values of roughly 8–16 Mg DM ha^{-1} yr^{-1}, consistent with the empirical range where 10-year rotations produced MAI of about 10–15 Mg DM ha^{-1} yr^{-1} for the best poplar cultivars in northern Poland and Italy. [11, 13]

In the agroforestry context, such age functions can be combined with tree density and allometric partitioning to above- and below-ground biomass to obtain tree-row biomass

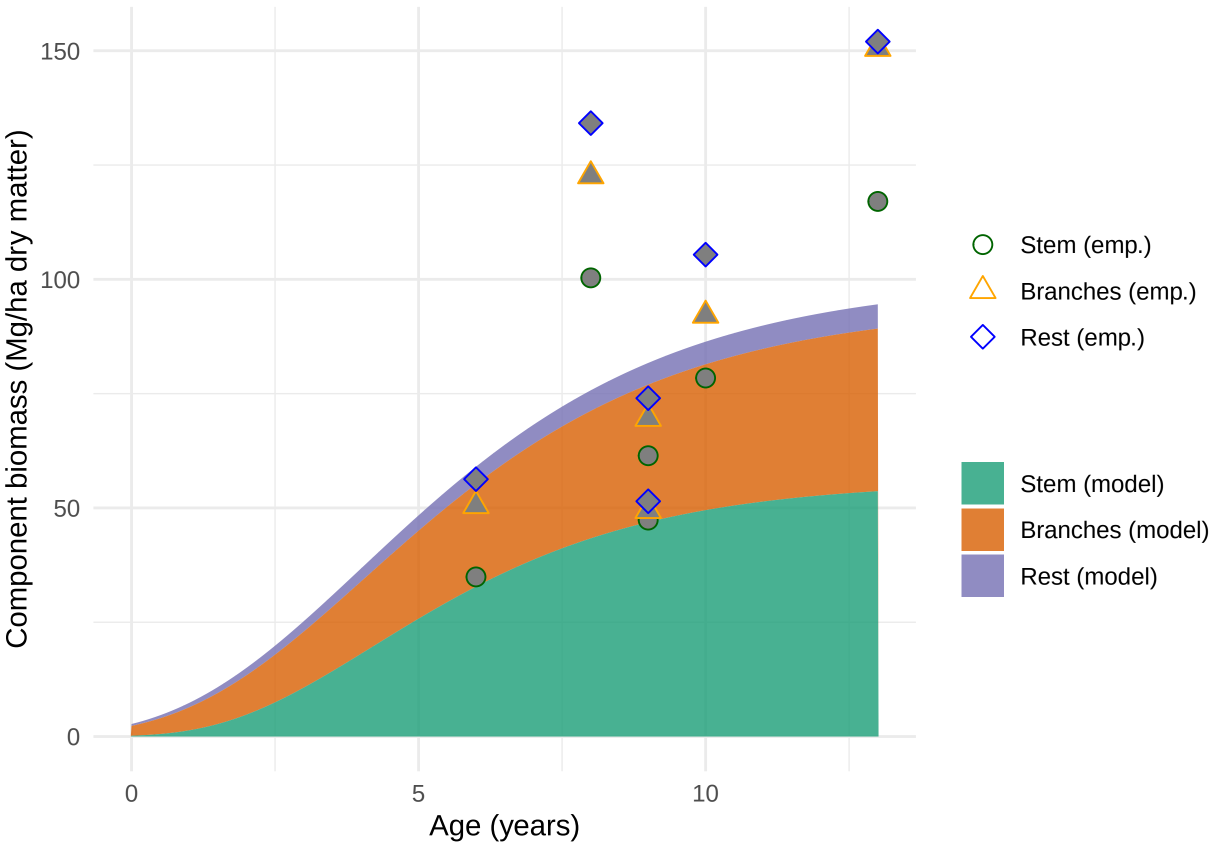

Figure 2: Modeled stand-level biomass components for poplar clone Max 3 over age, represented as stacked Gompertz curves and compared with empirical reference values from agroforestry trials.

over time for different designs. Calibration against local inventory data or growth measurements (e.g. basal area increment and height growth) is recommended before using the curves for scenario analysis.

2.5. Cascading use of biomass and product quality

From a cleaner production perspective, cascading use of biomass—prioritising high-value material uses before energy recovery—is widely advocated to maximise value creation and reduce environmental burdens Lamers et al. [8]; Espinoza et al. [5]. In wood-based value chains, this typically implies allocating higher-quality stem fractions to products such as sawn timber, veneer, panels, or pulp, while directing lower-quality fractions and residues to energy applications De Meyer et al. [2]. Multi-product MILP models such as

OPTIMASS implement this concept by defining a hierarchy of product quality classes and allowing higher-quality products to satisfy lower-quality demands via substitution constraints along the cascade De Meyer et al. [2]; De Meyer et al. [3].

For agroforestry biomass such as poplar and willow, this logic translates into a set of product streams (e.g. larger-diameter logs for material uses, medium-diameter fractions for pulp and panels, small-diameter wood and residues for energy), whose relative shares depend on stand age and growth conditions Huber et al. [7]; Trnka et al. [15]. In the present work, the age-class yield construction described above is extended by associating each age class with product-specific yield fractions, thereby capturing age-dependent quality in a simplified but optimisation-compatible form. The MILP formulation then enforces a product hierarchy in which higher-quality products can satisfy lower-quality demands, subject to the available yields and infrastructure capacities, thus embedding a simplified cascading logic into the regional supply chain design De Meyer et al. [2].

2.6. Synthesis and modelling gap

The literature summarised above provides three critical ingredients for modelling agroforestry biomass supply chains. First, empirical SRAFS and SRC studies offer species- and site-specific information on growth, MAI, stand structure, and economic performance, including allometric models that link diameter distributions to biomass Huber et al. [6]; Huber et al. [7]; Trnka et al. [15]; Testa et al. [13]. Second, age-dependent stand biomass functions such as Chapman–Richards provide a convenient parametric representation of growth over rotation, from which discrete age-class yields can be derived Niemczyk et al. [11]. Third, biomass supply chain MILP models and multi-product cascading formulations offer a rich toolkit for optimising infrastructure, logistics, and product allocation in regional bioeconomy systems Zahraee et al. [16]; De Meyer et al. [2]; De Meyer et al. [3]; Espinoza et al. [5].

At the same time, existing models fall short of explicitly coupling agroforestry stand dynamics with supply chain design in several respects. Most biomass supply chain formulations assume exogenous biomass availability profiles and do not represent stand age or rotation decisions, while forestry-oriented growth and yield models are rarely embedded in regional optimisation models that include multiple products and cascading uses De Meyer et al. [3]; Lamers et al. [8]. In particular, there is a lack of models that (i) link age-dependent stand biomass functions to age-class yield tables for different product types, (ii) enforce multi-cycle age-lag constraints that ensure consistency of harvesting decisions at each agroforestry site across successive rotations, and (iii) allocate biomass along a product quality cascade within a spatially explicit, multi-period supply chain optimisation framework De Meyer et al. [2]; De Meyer et al. [3]; Eksioglu et al. [4].

The MILP formulation developed in this paper addresses this gap by integrating age-dependent biomass yield modelling, stand-level age-lag constraints, and a product-cascading hierarchy into a single optimisation model for the design and operation of agroforestry biomass supply chains.

3. Biomass supply chain modelling

Biomass supply chain design and planning has been widely studied using MILP models at strategic, tactical, and operational levels [16, 12]. Comprehensive reviews classify models according to decision scope (location, capacity, routing, inventory), time horizon, uncertainty treatment, and sustainability metrics [16, 5]. Many models focus on bioenergy supply chains for electricity, heat, or biofuels, considering feedstocks such as agricultural residues, forestry residues, dedicated energy crops, and municipal wastes [12]. Environmental aspects, including greenhouse gas emissions and land use, are increasingly integrated, often using life-cycle assessment (LCA) data or emission factors [5].

Among generic models, OPTIMASS is a notable example of a multi-product MILP that optimises biomass supply chains with changing biomass characteristics and by-product re-injection [2]. It models the flow of multiple biomass types and products through supply chain stages, capturing conversion yields, storage, and cascading uses. The t-OPTIMASS extension further incorporates biomass growth and regeneration over multiple periods, enabling the representation of time-dependent availability for biomass supply chains [3]. However, these formulations generally assume exogenous biomass availability profiles per period, and do not explicitly model stand-level age dynamics or rotation constraints.

Several scenario-based approaches have been developed for forest biorefinery supply chains, linking product portfolios, process choices, and supply chain design under market volatility [9]. Lamers et al. [8] discuss strategic feedstock supply system design for biorefineries, emphasising the importance of economies of scale, conversion yields, and logistics in achieving cost-efficient and robust systems. Reviews highlight that only a limited subset of models explicitly consider biomass growth, spatially explicit land-use change, or agroforestry systems [16, 12]. There is therefore a clear opportunity to develop models that couple agroforestry stand dynamics with network design and cleaner production objectives.

3.1. Cascading use of biomass

From a cleaner production perspective, the cascading use of biomass—prioritising high-value material uses before energy recovery—is widely advocated to maximise resource efficiency and mitigate environmental impacts [8]. In wood-based value chains, this implies that high-quality timber or chemical feedstock fractions should first be allocated to material products such as engineered wood products, pulp and paper, or biochemicals, while lower-quality fractions and residues are used for energy [2, 8]. MILP models can represent such cascades by defining product quality classes and allowing substitution of higher-quality products for lower-quality demands, subject to hierarchy constraints.

The OPTIMASS framework explicitly models such product cascades by defining a product quality order and allowing flows from higher-quality products to lower-quality demand nodes [2]. Similar ideas have been applied in forest-based biorefinery supply chains, where product portfolios include platform chemicals, fuels, and electricity [9]. For agroforestry biomass, poplar and willow can supply different product streams, including sawlogs, peeled veneers, pulpwood, and energy chips, with quality primarily defined by diameter, straightness, and defect rates [13**?**]. Economic analyses in Italy and Central Europe suggest that the economic viability of SRC systems depends on access to higher-value markets such as panel products and pulp, in addition to energy markets [13].

Cleaner production strategies that integrate agroforestry biomass into regional bioeconomies therefore require models that can allocate biomass flows along such cascades while respecting stand growth dynamics, rotation constraints, and logistical capacities. Explicit representation of age-dependent yield quality (e.g., diameter distributions, wood density) remains challenging in large-scale MILP models, but can be approximated through age- and species-specific product yield fractions.

3.2. Stand-level yield functions and age classes

To integrate allometric growth into a MILP formulation, we abstract from individual trees and construct age-class yield functions for each biomass type p and age a. Following common practice in forest and SRC modelling, we define discrete annual age classes and compute the stand biomass yield Y_{pa} (t ha^{-1}) associated with harvesting at age a by combining allometric functions with empirically observed diameter distributions and stand densities [7, 15]. Where detailed stand structure data are available, this can be done by simulating SBD distributions over time (e.g., via mixed-effects models) and applying tree-level allometrics to each diameter class [7]. In the absence of such detailed data, age-dependent MAI and CAI curves reported in the literature can be used to approximate Y_{pa} [15, 7].

In the context of the agroforestry supply chain design (SCD) model, we assume that each agroforestry site i is characterised by a species or species mix, a planting density, and management regime, which jointly determine a site-specific yield profile η_{pa} (t ha^{-1}) for each product type p at age a. These yield profiles can be interpreted as the maximum harvestable biomass per hectare for product p if the stand is harvested at age a, conditional on the stand having been established at an earlier time. In the current MILP formulation, these profiles are captured through the parameter η_{pa}, termed base yield, and are linked to decision variables via the constraint

$$Y_{ipt} = \sum_{s=1}^{t-1} \eta_{p(t-s)} \cdot \mathrm{AREA}_i \cdot z_{ist} \quad \forall i, p, t \in \mathcal{T}^{\mathrm{harv}}.$$

Here, Y_{ipt} is the maximum harvest quantity of product p at site i in period t, AREA_i is the site area (ha), and z_{ist} is a binary variable indicating that harvests at site i in periods s and t form consecutive harvests with an age lag $t-s$ within allowed bounds. The term $\eta_{p(t-s)}$ thus plays the role of an allometrically derived yield coefficient that depends only on stand age $a = t - s$, not on calendar time.

3.3. Biomass types and product quality classes

In the model, three product types are distinguished: $p = 1$ (chemical-grade biomass), $p = 2$ (pulp-grade biomass), and $p = 3$ (energy-grade biomass). This tripartite classification reflects a product quality hierarchy, with product 1 having the highest quality and price, and product 3 being the lowest-quality fraction typically used for energy [2, 8]. From an allometric perspective, the differentiation among these products can be linked to diameter classes, stem straightness, and defect rates, as these determine the suitability of biomass fractions for different processing routes [13**?**].

In practice, allometric models provide estimates of total aboveground woody biomass, which can be split into product classes using empirical conversion factors or allocation rules derived from product recovery studies (e.g., share of biomass suitable for veneer vs. pulp vs. chips as a function of diameter and log length) [**?**]. For the purposes of the SCD model, we can represent this by specifying product-specific yield coefficients η_{pa} that sum to the total biomass yield at age a and reflect the expected proportion of each product class under a given management regime.

The cascading hierarchy in the MILP formulation is implemented through demand satisfaction constraints that allow higher-quality products to satisfy lower-quality demands. Specifically, demand $D^{\max}_{kpt}$ at consumer k for product class p in period t can be met by flows of product p and all higher-quality classes $p' \leq p$ from storage hubs, as captured by

$$\sum_{j \in \mathcal{J}, p' \in \mathcal{P} : p' \leq p} X_{jkp'pt} \leq D^{\max}_{kpt} \quad \forall k, p, t.$$

This structure supports cleaner production by favouring material uses when possible and relegating residual fractions to energy uses, depending on revenue parameters $R_{k,p}$ and system constraints [2, 8].

4. MILP model for agroforestry supply chain design

4.1. Sets, indices, and time structure

The model considers a set of agroforestry sites $\mathcal{I}$, storage/processing facilities $\mathcal{J}$, consumer sites $\mathcal{K}$, and product types $\mathcal{P} = \{1, 2, 3\}$. Time is discretised into annual periods $t \in \mathcal{T} = \{1, \ldots, T_{\max}\}$. To represent stand establishment and harvest timing, extended time sets $\mathcal{T}^{+} = \{0, \ldots, T_{\max} + 1\}$ and harvest periods $\mathcal{T}^{\text{harv}} = \{A^{\min} + 1, \ldots, T_{\max}\}$ are defined, where $A^{\min}$ and $A^{\max}$ are minimum and maximum stand ages (years) at harvest. The set $\mathcal{S}$ comprises all ordered pairs (s, t) of consecutive harvest periods (including establishment at $s = 0$) satisfying the age-lag bounds:

$$\mathcal{S} = \{(s, t) \mid s, t \in \{0, \ldots, T_{\max} + 1\},\ t - s \in \{A^{\min}, \ldots, A^{\max}\}\}.$$

4.2. Decision variables

Binary variables z_{ist} indicate whether, at site i, periods s and t form consecutive harvests (or establishment and first harvest if $s = 0$), with $(s, t) \in \mathcal{S}$. These variables define a path of consecutive harvest cycles at each site over the planning horizon. Continuous variables $Y_{ipt} \geq 0$ represent the maximum harvest quantity (t) of product p at site i in harvest period t. Flow variables X_{ijpt} and $X_{jkpp't}$ represent transport of products from sites to storage and from storage to consumers, respectively, and S_{jpt} represent inventories at storage locations.

4.3. Objective function

The model maximises net present value over the planning horizon, expressed as total revenues minus establishment, opportunity, harvest, transport, and storage costs:

$$\begin{aligned}
\max Z = & \sum_{k,p,t} R_{k,p} \cdot \sum_{j \in \mathcal{J},\, p' \in \mathcal{P}: p' \leq p} X_{jkp'pt} \\
& - \sum_{i} C_i^{\text{est}} \cdot \text{AREA}_i \cdot \sum_{t \in \mathcal{T}} z_{i0t} \\
& - \sum_{i} C_i^{\text{opp}} \cdot \text{AREA}_i \cdot \sum_{t \in \mathcal{T}} (T_{\max} - t) \cdot z_{i0t} \\
& - \sum_{i,(s,t) \in \mathcal{S}: s > 0} C_i^{\text{harv}} \cdot \text{AREA}_i \cdot z_{ist} \\
& - c^{\text{tr-raw}} \sum_{i,j,p,t} d_{ij} \cdot X_{ijpt} - c^{\text{tr-pre}} \sum_{j,k,p,t} d_{jk} \cdot X_{jkpt} - \sum_{j,p,t} c_j^{\text{stor}} \cdot S_{jpt}.
\end{aligned}$$

Here, $R_{k,p}$ are product- and consumer-specific revenues (€/t), C_i^{est} are establishment costs per hectare, C_i^{opp} are annual opportunity costs per hectare, and C_i^{harv} are harvest and refitting costs per hectare [13]. Transport costs are proportional to flow and distance with rates $c^{\text{tr-raw}}$ and $c^{\text{tr-pre}}$ for raw and pre-treated biomass, respectively, and storage incurs linear holding costs c_j^{stor} per tonne.

4.4. Key constraints

The path connectivity constraint ensures that each harvest period at a site has exactly one predecessor and one successor in the age-lag graph, effectively forming a path over $\mathcal{T}^+$:

$$\sum_{s=0:(s,t)\in\mathcal{S}}^{t} z_{ist} = \sum_{u=t+1:(t,u)\in\mathcal{S}}^{T_{\max}+1} z_{itu} \quad \forall i \in \mathcal{I}, t \in \mathcal{T}.$$

Establishment is restricted to at most one time per site:

$$\sum_{t\in\mathcal{T}^+} z_{i0t} \leq 1 \quad \forall i \in \mathcal{I}.$$

Biomass yield is linked to age-lag decisions through the allometric yield coefficients:

$$Y_{ipt} = \sum_{s=1}^{t-1} \eta_{p(t-s)} \cdot \mathrm{AREA}_i \cdot z_{ist} \quad \forall i \in \mathcal{I}, p \in \mathcal{P}, t \in \mathcal{T}^{\mathrm{harv}}.$$

Harvest quantities bound outbound flows from each site:

$$\sum_{j\in\mathcal{J}} X_{ijpt} \leq Y_{ipt} \quad \forall i, p, t \in \mathcal{T}^{\mathrm{harv}}.$$

At storage hubs, inventories obey standard balance equations with initial inventory zero:

$$S_{jpt} = S_{jp,t-1} + \sum_{i} X_{ijpt} - \sum_{k,p'\geq p} X_{jkpp't} \quad \forall j, p, t \in \mathcal{T}^{\mathrm{harv}}.$$

Storage and processing capacities are enforced via:

$$\sum_{p} S_{jpt} \leq \mathrm{CAP}_j^{\mathrm{stor}} \quad \forall j, t, \qquad \sum_{i,p} X_{ijpt} \leq \mathrm{CAP}_j^{\mathrm{proc}} \quad \forall j, t.$$

Demand satisfaction with cascades is modelled as described above, where higher-quality products can satisfy lower-quality demands up to maximum demand levels $D_{kpt}^{\max}$.

5. Case study and computational experiment plan

The proposed MILP model is intended to support the design of agroforestry biomass supply chains for cleaner production in specific regional contexts. In a forthcoming case study, we focus on poplar-based alley-cropping systems in a temperate European setting, informed by empirical data from SRAFS trials in Southern Germany and SRC plantations in Central Europe [7, 15, 13].

We envisage the following steps and experimental scenarios for computational analysis:

1. **Parameterisation of allometric yield profiles:** Use species- and site-specific allometric models and observed growth trajectories to construct age-dependent yield coefficients η_{pa} for chemical, pulp, and energy product classes at representative agroforestry sites. Where necessary, scenario ranges will be defined to capture uncertainty in MAI and CAI due to climate variability and management differences [6, 7, 15].
2. **Baseline optimisation:** Solve the MILP model for a baseline configuration with fixed prices, capacities, and policy settings, obtaining optimal establishment timing, harvest schedules, and supply chain structure (locations, flows, inventories).
3. **Rotation length scenarios:** Vary the age-lag bounds $A^{\min}$ and $A^{\max}$ to represent alternative rotation strategies (e.g., 3–5 years vs. 8–10 years) and analyse their impacts on profitability, product mix, and land use. This will shed light on trade-offs between shorter rotations favouring energy chips and longer rotations enabling higher shares of material products [15, 13].
4. **Product price and cascading scenarios:** Explore scenarios with different relative prices for chemical, pulp, and energy products, reflecting changes in market demand and policy incentives (e.g., bioenergy subsidies, green chemicals markets). Assess how the product cascade hierarchy and yield quality structure influence the marginal value of species and rotation decisions [2, 8].
5. **Infrastructure and logistics scenarios:** Evaluate the effect of alternative storage and processing network configurations (e.g., decentralised vs. centralised hubs, different capacity levels, transport cost parameters) on the optimal spatial deployment of agroforestry sites and biomass flows [2, 3, 8].
6. **Sensitivity to allometric and growth uncertainty:** Conduct parametric sensitivity and, where computationally feasible, stochastic analyses to quantify the impact of uncertainty in allometric parameters and growth responses (e.g., under drought or management differences) on optimal decisions and system robustness [7, 15].
7. **Cleaner production indicators:** For selected solutions, compute derived indica-

tors such as biomass cascading index (share of biomass used for material products before energy), utilisation rates of higher-quality fractions, and, where coupled with LCA data, greenhouse gas emission savings relative to fossil benchmarks [5, 16].

6. Conclusions

This paper outlines a research framework that couples allometric biomass growth models with a MILP-based agroforestry biomass supply chain design model featuring age-lag constraints and product cascading. Building on empirical evidence from temperate SRAFS and SRC systems and established biomass supply chain models, the proposed approach aims to capture key interactions between stand-level growth dynamics and network-level design decisions that are central to the cleaner production potential of agroforestry [7, 2, 3]. Future work will implement and calibrate the model for a regional case study, conduct the computational experiments sketched above, and extend the framework towards multi-objective optimisation including environmental indicators such as greenhouse gas emissions and biodiversity proxies.

References

[1] Mahsa Arabi, Saeed Yaghoubi, and Javad Tajik. A mathematical model for microalgae-based biobutanol supply chain network design under harvesting and drying uncertainties. *Energy*, 179: 1004–1016, 2019. ISSN 0360-5442. doi: 10.1016/j.energy.2019.04.219.

[2] Annelies De Meyer, Dirk Cattrysse, Jussi Rasinmäki, and Jos Van Orshoven. A generic mathematical model to optimise strategic and tactical decisions in biomass-based supply chains (optimass). *European Journal of Operational Research*, 245(1):247–264, 2015. ISSN 0377-2217. doi: 10.1016/j.ejor.2015.02.042. URL `https://www.sciencedirect.com/science/article/pii/S0377221715001654`.

[3] Annelies De Meyer, Dirk Cattrysse, Jussi Rasinmäki, and Jos Van Orshoven. Considering biomass growth and regeneration in the optimisation of biomass supply chains. *Renewable Energy*, 87:990–1002, 2016. ISSN 0960-1481. doi: 10.1016/j.renene.2015.09.038. URL `https://www.sciencedirect.com/science/article/pii/S0960148115301397`.

[4] Sandra D. Eksioglu, Ambarish Acharya, Liam E. Leightley, and Sumesh Arora. Analyzing the design and management of biomass-to-biorefinery supply chain. *Computers & Industrial Engineering*, 57 (4):1342–1352, 2009. ISSN 0360-8352. doi: 10.1016/j.cie.2009.07.003.

[5] Diego Espinoza, Farid Chejne, José M. Ponce-Ortega, and Francisco Nápoles-Rivera. Sustainable biorefinery supply chain design and management: A review. *Bioresource Technology*, 243:1223–1235, 2017. ISSN 0960-8524. doi: 10.1016/j.biortech.2017.07.089.

[6] Julia A. Huber, Katharina May, and Kurt-Jürgen Hülsbergen. Allometric tree biomass models of various species grown in short-rotation agroforestry systems. *European Journal of Forest Research*, 135(2):283–296, 2016. ISSN 1612-4669. doi: 10.1007/s10342-015-0932-5.

[7] Julia A. Huber, Michael Matiu, and Kurt-Jürgen Hülsbergen. First-rotation growth and stand structure dynamics of tree species in organic and conventional short-rotation agroforestry systems. *Heliyon*, 4(7):e00645, 2018. ISSN 2405-8440. doi: 10.1016/j.heliyon.2018.e00645. URL `https://www.sciencedirect.com/science/article/pii/S2405844017334904`.

[8] Patrick Lamers, Eric Tan, Erin Searcy, Christopher Scarlata, Kara Cafferty, and Jacob Jacobson. Strategic supply system design – a holistic evaluation of operational and production cost for a biorefinery supply chain. *Biofuels, Bioproducts and Biorefining*, 9(6):648–660, 2015.

[9] Behzad Mansoornejad, Virginie Chambost, and Paul Stuart. Multi-period optimization of forest biorefinery supply chains integrating bioenergy and biofuels production. *Chemical Engineering Research and Design*, 91(8):1454–1468, 2013. ISSN 0263-8762. doi: 10.1016/j.cherd.2013.02.009.

[10] P. K. Ramachandran Nair. *An Introduction to Agroforestry*. Kluwer Academic Publishers, Dordrecht, 1993. ISBN 978-0792321347.

[11] Marzena Niemczyk, Mariusz J. Stolarski, Marta Krzyżaniak, et al. The effects of cultivar and rotation length (5 vs. 10 years) on biomass production and sustainability of poplar (populus spp.) bioenergy plantation. *GCB Bioenergy*, 13:999–1014, 2021. doi: 10.1111/gcbb.12827.

[12] B. Sharma, R. G. Ingalls, C. L. Jones, and A. Khanchi. A review of biomass supply chain management studies in the context of bioenergy production. *Biomass and Bioenergy*, 56:147–158, 2013. ISSN 0961-9534. doi: 10.1016/j.biombioe.2013.04.005.

[13] Riccardo Testa, Anna Maria Di Trapani, Mario Foderà, Filippo Sgroi, and Salvatore Tudisca. Economic evaluation of introduction of poplar as biomass crop in italy. *Renewable and Sustainable Energy Reviews*, 38:775–780, 2014. ISSN 1364-0321. doi: 10.1016/j.rser.2014.07.054. URL `https://www.sciencedirect.com/science/article/pii/S1364032114005473`.

[14] Eric Toensmeier. Perennial staple crops and agroforestry for climate change mitigation. In Florencia Montagnini, editor, *Integrating Landscapes: Agroforestry for Biodiversity Conservation and Food Sovereignty*, volume 12 of *Advances in Agroforestry*, pages 431–453. Springer, Cham, 2017. doi: 10.1007/978-3-319-69371-2_18.

[15] Miroslav Trnka, J. Fialová, V. Koutecký, M. Fajman, Z. Šalud, and S. Hejduk. Biomass production and survival rates of selected poplar clones grown under a short-rotation system on arable land. *Plant, Soil and Environment*, 54(2):78–88, 2008. ISSN 1214-1178. URL `https://www.agriculturejournals.cz/web/pse.htm?type=article&id=419_2007-PSE`.

[16] Seyed Mohammad Zahraee, Bijan Bahrami, Mehdi Baghery, and Arash Ahmadi. A comprehensive review of biomass supply chain modelling and optimisation. *Renewable and Sustainable Energy Reviews*, 110:1–19, 2019. ISSN 1364-0321. doi: 10.1016/j.rser.2019.04.071.